

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients**, processing **10,000+ monthly submissions** via REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack data platform for the AI team (React, Node.js, MongoDB), cutting manual work by **100+ hours/month** and lifting labeling throughput by **60%**.
- Built backend reliability and security controls, AWS KMS encryption, auth hardening, rate limiting, and integration tests, remediating **20+ VAPT findings**.
- Shipped features rapidly to production in a small team as primary technical POC, mentored a second developer, and piloted a COO-sponsored GitHub Copilot experiment.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with input validation and clear contracts, used in production by the frontend team in a fast-moving startup.
- Migrated core repositories from JavaScript to TypeScript and shipped a production LLM feature (**4-bit quantized Llama-7B via llama.cpp**) with async queued generation and content-safety filtering.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform independently end to end; live **3+** years.

Projects

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Optimized a high-performance Spark data platform over a **~1.4B-row** dataset with broadcast joins, adaptive execution, skew-join handling, partitioning, and column pruning, cutting end-to-end runtime **~25%** and the analytics tail-latency ratio from **2.14x to 1.74x**, profiled via Spark UI.

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI research platform that turns open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with cell-level provenance; hardened with **150+ tests** and deployed as a live cloud service.

ngvi-curvature-variance – Natural-Gradient VI Study (GitHub) Python, PyTorch, NumPy

- Ran a matched-compute study of natural-gradient variational inference vs Adam on ill-conditioned posteriors (**PyTorch**, black-box VI), grounded in probability and Fisher-information geometry, showing NGVI's ELBO gain grows with curvature.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, C++, Java, TypeScript, JavaScript, SQL, Bash

Performance: Spark / PySpark, distributed computing, C++/SIMD, Linux perf, caching, optimization

Backend & Web: FastAPI, Node.js, Express, React, Next.js, REST APIs

ML & Statistics: PyTorch, NumPy, probability & statistics, NLP, variational inference, evaluation

Data & Cloud: MongoDB, PostgreSQL, ETL pipelines, AWS, GCP, Azure, Docker, Linux

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.